



DEPARTMENT OF MATHEMATICS AND STATISTICS
SEMESTER I ACADEMIC YEAR 2025/2026
MS 8221: DIFFERENTIAL EQUATIONS
UQF 8 THIRD YEAR: COMPUTER ENGINEERING
LECTURE 2
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2.1.2: A Second Approach to Power Series Solution of Differential Equations/A General Approach to Power Series Solution

CONCEPTS:

- 1) The method developed in the previous section works satisfactorily if only the first few terms in a power series solution are required, but it has the disadvantage that a separate calculation is required each time a coefficient is determined.
- 2) The present section shows how in many cases this difficulty can be overcome by introducing a systematic and simple way of generating arbitrarily many terms in a power series solution.
- 3) For a given ODE $y'' + p(x)y' + q(x)y = 0$, we first represent $p(x)$ and $q(x)$ by power series in powers of x (or powers of $(x - x_0)$ if solutions in powers of $x - x_0$ are wanted).
 $\Rightarrow$ Often $p(x)$ and $q(x)$ are polynomials, and then nothing needs to be done in this step.
- 4) Next, we assume a solution in the form of a **Power Series** with unknown coefficients a_n as:

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_{n-1}x^{n-1} + a_nx^n + a_{n+1}x^{n+1} + \cdots = \sum_{n=0}^{\infty} a_nx^n$$

and insert this series and the series obtained by termwise differentiation,

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots + na_nx^{n-1} + (n+1)a_{n+1}x^n + (n+2)a_{n+2}x^{n+1} + \cdots$$

$$\Rightarrow y'(x) = \sum_{n=1}^{\infty} na_nx^{n-1}$$

and

$$y''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \cdots + (n+1)na_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \cdots$$

$$\Rightarrow y''(x) = \sum_{n=2}^{\infty} n(n-1)a_nx^{n-2}$$

into the ODE.

- 5) The approach enables the coefficients a_n of the power series solution to be determined by means of a **Recurrence Formula/Recurrence Relation/Recursion Formula**.

2.1.2.1: Recurrence Formula by Using Individual Terms

CONCEPTS:

- 1) A second order linear differential equation

$$b_2(x)y'' + b_1(x)y' + b_0y = g(x) \quad (1)$$

has variable coefficients when $b_2(x)$, $b_1(x)$ and $b_0(x)$ are not all constants.

$\Rightarrow$ If $b_2(x) \neq 0$, then we can divide by it and rewrite equation (1) as:

$$y'' + p(x)y' + q(x)y = r(x) \quad (2)$$

$$\Rightarrow \text{Where, } p(x) = \frac{b_1(x)}{b_2(x)}, \quad q(x) = \frac{b_0(x)}{b_2(x)} \text{ and } r(x) = \frac{g(x)}{b_2(x)}.$$

- 2) Equation (1) is called **Homogeneous** when $g(x) = 0$, in which case equation (2) specializes to:

$$y'' + p(x)y' + q(x)y = 0 \quad (3)$$

- 3) **THEOREM:**

$\Rightarrow$ If $x = 0$ is an ordinary point of equation (3), then the general solution of equation (3) has the form:

$$y(x) = \sum_{n=0}^{\infty} a_nx^n = a_0y_1(x) + a_1y_2(x) \quad (4)$$

where a_0 and a_1 are arbitrary constants, and $y_1(x)$ and $y_2(x)$ are **Linearly Independent Functions** analytic at $x = 0$.

- 4) To evaluate the coefficients a_n in the solution furnished by the Theorem above, we use the following **Five-Step Procedure** known as **Power Series Method**.

STEP 1:

⇒ Substitute into the left side of the homogeneous differential equation the power series

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_{n-1}x^{n-1} + a_nx^n + a_{n+1}x^{n+1} + \cdots \quad (5)$$

together with the power series for

$$y' = a_1 + 2a_2x + 3a_3x^2 + \cdots + na_nx^{n-1} + (n+1)a_{n+1}x^n + (n+2)a_{n+2}x^{n+1} + \cdots \quad (6)$$

and

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + \cdots + (n+1)na_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \cdots \quad (7)$$

STEP 2:

⇒ Collect powers of x and set the coefficients of each power of x equal to zero.

STEP 3:

⇒ The equation obtained by setting the coefficients of x^n to zero in Step 2 will contain a_j terms for a finite number of j values.

⇒ Solve this equation for the a_j term having the **Largest Subscript**. The resulting equation is known as the **Recurrence Formula** for the differential equation.

STEP 4:

⇒ Use the recurrence formula to sequentially determine a_j ($j = 2, 3, 4, \dots$) in terms of a_0 and a_1 .

STEP 5:

⇒ Substitute the coefficients determined in Step 4 into equation (5) and rewrite the solution in the form of equation (4).

- 5) The power series method is only applicable when $x = 0$ is an ordinary point, although a differential equation must be in the form of equation (2) to determine whether $x = 0$ is an ordinary point. Once this condition is verified, the power series method can be used on either form (1) or (2).
- 6) If $g(x)$ in (1) is not zero, then the equation is called **Nonhomogeneous Equation**. If $r(x)$ in (2) is analytic at $x = 0$, it has a **Taylor Series Expansion** around that point and the power series method given above can be modified.
 - ⇒ In Step 1, equation (5) through (7) are substituted into the left side of the nonhomogeneous equation; the right side is written as a Taylor Series around the origin.
 - ⇒ Steps 2 and 3 change so that the coefficients of each power of x on the left side of the equation resulting from Step 1 are set equal to their counterparts on the right side of that equation.
 - ⇒ The form of the solution in Step 5 becomes:

$$y(x) = a_0y_1(x) + a_1y_2(x) + y_3(x)$$

- 7) When solutions are required around the ordinary point $x_0 \neq 0$, it is generally simplifies the algebra if x_0 is translated to the origin by the **Change of Variables**:

$$t = x - x_0$$

⇒ The solution of the new differential equation that results can be obtained by the power series method about $t = 0$.

⇒ Then the solution of the original equation is easily gotten by **Back-Substitution**.

⇒ That is, from $t = x - x_0$, we have;

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt}(1) = \frac{dy}{dt}$$

and

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dt} \right) \\ \Rightarrow \frac{d^2y}{dx^2} &= \frac{d}{dt} \left(\frac{dy}{dt} \right) \frac{dt}{dx} = \frac{d^2y}{dt^2}(1) = \frac{d^2y}{dt^2} \end{aligned}$$

EXAMPLES

1. (a) Show that $x = 0$ is an ordinary point of:

$$y'' - xy' + 2y = 0$$

- (b) Find a recurrence formula for the power series solution around $x = 0$ for the differential equation given in 1. (a) above.
(c) Find the general solution near $x = 0$ of:

$$y'' - xy' + 2y = 0$$

SOLUTION

- (a) From $y'' - xy' + 2y = 0$, we have $p(x) = -x$ and $q(x) = 2$ which are both polynomials, hence they are analytic everywhere.
⇒ Therefore, every value of x is an ordinary point of the given equation, in particular $x = 0$ is an ordinary point.
(b) It follows from problem 1.(a) above that $x = 0$ is an ordinary point of the given equation, so substituting equation (5) through (7) into the left side of the differential equation, we get;

$$\begin{aligned} &[2a_2 + 6a_3x + 12a_4x^2 + \cdots + (n+1)(n)a_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \cdots] \\ &\quad -x[a_1 + 2a_2x + 3a_3x^2 + \cdots + na_nx^{n-1} + (n+1)a_{n+1}x^n + \cdots] \\ &\quad + 2[a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_{n-1}x^{n-1} + a_nx^n + a_{n+1}x^{n+1} + \cdots] = 0 \end{aligned}$$

⇒ Combining terms that contain like powers of x , we have;

$$(2a_2 + 2a_0) + (6a_3 + a_1)x + (12a_4)x^2 + \cdots + [(n+2)(n+1)a_{n+2} - na_n + 2a_n]x^n + \cdots = 0 + 0x + 0x^2 + \cdots + 0x^n + \cdots$$

⇒ This equation holds if and only if each coefficient in the left side is zero. Thus,
 $2a_2 + 2a_0 = 0, \quad 6a_3 + a_1 = 0, \quad 12a_4 = 0, \quad \cdots$

⇒ In general:

$$\begin{aligned} (n+2)(n+1)a_{n+2} - (n-2)a_n &= 0 \\ \Rightarrow a_{n+2} &= \frac{(n-2)}{(n+2)(n+1)}a_n; \quad \text{for } n = 0, 1, 2, \cdots \end{aligned}$$

$\Rightarrow$ The equation $a_{n+2} = \frac{(n-2)}{(n+2)(n+1)}a_n$ is the **Recurrence Formula** for this problem.

- (c) Successively evaluating the recurrence formula obtained in part (b) above for $n = 0, 1, 2, 3, \dots$, we calculate:

$$\begin{aligned} a_2 &= -a_0, \\ a_3 &= -\frac{1}{6}a_1, \\ a_4 &= 0, \\ a_5 &= \frac{1}{20}a_3 = \frac{1}{20}\left(-\frac{1}{6}a_1\right) = -\frac{1}{120}a_1, \\ a_6 &= \frac{2}{30}a_4 = \frac{1}{15}(0) = 0, \\ a_7 &= \frac{3}{42}a_5 = \frac{1}{14}\left(-\frac{1}{120}a_1\right) = -\frac{1}{1680}a_1, \\ a_8 &= \frac{4}{56}a_6 = \frac{1}{14}(0) = 0, \quad \dots \end{aligned}$$

$\Rightarrow$ Note that since $a_4 = 0$, it follows from the recurrence formula that all the even coefficients beyond a_4 are also zero.

$\Rightarrow$ Substituting these values into equation (5), we have;

$$\begin{aligned} y(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \\ y(x) &= a_0 + a_1x - a_0x^2 - \frac{1}{6}a_1x^3 + 0x^4 - \frac{1}{120}a_1x^5 + 0x^6 - \frac{1}{1680}a_1x^7 + 0x^8 - \dots \\ y(x) &= a_0(1 - x^2) + a_1\left(x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \frac{1}{1680}x^7 - \dots\right) \end{aligned}$$

$\Rightarrow$ **Note that:** If we define:

$$\begin{aligned} y_1(x) &= 1 - x^2 \quad \text{and} \\ y_2(x) &= x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \frac{1}{1680}x^7 - \dots \end{aligned}$$

then the general solution can be rewritten as:

$$y(x) = a_0y_1(x) + a_1y_2(x)$$

2. Find a recurrence formula in the power series solution around $x = 0$ for the differential equation $y'' + y = 0$, hence find the general solution near $x = 0$.

SOLUTION

$\Rightarrow$ From $y'' + y = 0$, we have $p(x) = 0$ and $q(x) = 1$ which are both constants, hence they are analytic everywhere. Thus, $x = 0$ is an ordinary point of the given equation. Hence we have;

$$\begin{aligned} [2a_2 + 6a_3x + 12a_4x^2 + \dots + (n+1)(n)a_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \dots] \\ + [a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{n-1}x^{n-1} + a_nx^n + a_{n+1}x^{n+1} + \dots] = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow (2a_2 + a_0) + (6a_3 + a_1)x + (12a_4 + a_2)x^2 + \dots + [(n+2)(n+1)a_{n+2} + a_n]x^n + \dots = \\ 0 + 0x + 0x^2 + \dots + 0x^n + \dots \end{aligned}$$

⇒ Equating each coefficient to zero, we have;

$$2a_2 + a_0 = 0, \quad 6a_3 + a_1 = 0, \quad 12a_4 + a_2 = 0, \quad \dots$$

⇒ In general:

$$(n+2)(n+1)a_{n+2} + a_n = 0$$

$$\Rightarrow a_{n+2} = -\frac{1}{(n+2)(n+1)}a_n; \quad \text{for } n = 0, 1, 2, \dots$$

⇒ the equation $a_{n+2} = -\frac{1}{(n+2)(n+1)}a_n$ is the recurrence formula for this problem. Solving by power series method, we successively evaluate the recurrence formula for $n = 0, 1, 2, 3, \dots$, obtaining:

$$\begin{aligned} a_2 &= -\frac{1}{2}a_0 = -\frac{1}{2!}a_0, \\ a_3 &= -\frac{1}{6}a_1 = -\frac{1}{3!}a_1, \\ a_4 &= -\frac{1}{12}a_2 = -\frac{1}{12}\left(-\frac{1}{2}a_0\right) = \frac{1}{24}a_0 = \frac{1}{4!}a_0, \\ a_5 &= -\frac{1}{20}a_3 = -\frac{1}{20}a_3\left(-\frac{1}{6}a_1\right) = \frac{1}{120}a_1 = \frac{1}{5!}a_1, \\ a_6 &= -\frac{1}{30}a_4 = -\frac{1}{30}\left(\frac{1}{24}a_0\right) = -\frac{1}{720}a_0 = -\frac{1}{6!}a_0, \\ a_7 &= -\frac{1}{42}a_5 = -\frac{1}{42}\left(\frac{1}{120}a_1\right) = -\frac{1}{5040}a_1 = -\frac{1}{7!}a_1, \quad \dots \end{aligned}$$

$$\begin{aligned} y(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots \\ &= a_0 + a_1x - \frac{1}{2!}a_0x^2 - \frac{1}{3!}a_1x^3 + \frac{1}{4!}a_0x^4 + \frac{1}{5!}a_1x^5 - \frac{1}{6!}a_0x^6 - \frac{1}{7!}a_1x^7 + \dots \\ \Rightarrow y(x) &= a_0 \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) + a_1 \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots\right) \\ \Rightarrow \text{But } \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right) &= \cos x \text{ and} \\ \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots\right) &= \sin x \end{aligned}$$

⇒ Therefore, $y(x) = a_0 \cos x + a_1 \sin x$, where a_0 and a_1 are the arbitrary constants.

3. Find a recurrence formula in the power series solution around $x = 0$ for the nonhomogeneous differential equation:

$$(x^2 + 4)y'' + xy = x + 2$$

Hence find its general solution near $x = 0$.

SOLUTION

⇒ Dividing the given equation by $(x^2 + 4)$, we see that $x = 0$ is an ordinary point and that $r(x) = \frac{(x+2)}{(x^2+4)}$ is analytic there.

$\Rightarrow$ Thus, the power series method is applicable to the entire equation, which, furthermore, we may leave it in the form originally given to simplify the algebra. Thus we have;

$$(x^2 + 4)[2a_2 + 6a_3x + \cdots + n(n-1)a_nx^{n-2} + (n+1)na_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \cdots] \\ + x[a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + a_nx^n + \cdots] = x + 2$$

$$\Rightarrow x^2[2a_2 + 6a_3x + \cdots + n(n-1)a_nx^{n-2} + \cdots] \\ + 4[2a_2 + 6a_3x + 12a_4x^2 + \cdots + (n+2)(n+1)a_{n+2}x^n + \cdots] \\ + x[a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1} + \cdots] = x + 2$$

$$\Rightarrow (8a_2) + (24a_3 + a_0)x + (2a_2 + 48a_4 + a_1)x^2 + \cdots \\ + [n(n+1)a_n + 4(n+2)(n+1)a_{n+2} + a_{n-1}]x^n + \cdots = 2 + (1)x + 0x^2 + \cdots + 0x^n + \cdots \quad (*)$$

$\Rightarrow$ Equating coefficients of like powers of x , we have;

$$8a_2 = 2, \quad 24a_3 + a_0 = 1, \quad 2a_2 + 48a_4 + a_1 = 0, \cdots \quad (**)$$

$\Rightarrow$ In general:

$$n(n+1)a_n + 4(n+2)(n+1)a_{n+2} + a_{n-1} = 0; \quad \text{for } n = 2, 3, 4, \cdots \\ \Rightarrow a_{n+2} = -\frac{n(n-1)}{4(n+2)(n+1)}a_n - \frac{1}{4(n+2)(n+1)}a_{n-1}; \quad \text{for } n = 2, 3, 4, \cdots$$

$\Rightarrow$ Note that the recurrence formula is not valid for $n = 0$ and $n = 1$, since the coefficients of x^0 and x^1 on the right side of $(*)$ are not zero.

$\Rightarrow$ Instead, we use the first two equations in $(**)$ to obtain:

$$a_2 = \frac{1}{4} \quad \text{and} \quad a_3 = \frac{1}{24} - \frac{1}{24}a_0 \quad (***)$$

$\Rightarrow$ Using the results above, we have that a_2 and a_3 are given by $(***)$ and $a_4, a_5, a_6, \cdots$ for $n = 2, 3, 4, \cdots$ are given by the recurrence formula.

$\Rightarrow$ Thus, it follows from the recurrence formula that:

$$a_4 = -\frac{1}{24}a_2 - \frac{1}{48}a_1 = -\frac{1}{24}\left(\frac{1}{4}\right) - \frac{1}{48}a_1 = -\frac{1}{96} - \frac{1}{48}a_1, \\ a_5 = -\frac{3}{40}a_3 - \frac{1}{80}a_2 = -\frac{3}{40}\left(\frac{1}{24} - \frac{1}{24}a_0\right) - \frac{1}{80}\left(\frac{1}{4}\right) = -\frac{1}{160} + \frac{1}{320}a_0, \cdots$$

$\Rightarrow$ Thus, we have;

$$y(x) = a_0 + a_1x + \frac{1}{4}x^2 + \left(\frac{1}{24} - \frac{1}{24}a_0\right)x^3 + \left(-\frac{1}{96} - \frac{1}{48}a_1\right)x^4 + \left(-\frac{1}{160} + \frac{1}{320}a_0\right)x^5 + \cdots \\ \Rightarrow y(x) = a_0 \left(1 - \frac{1}{24}x^3 + \frac{1}{320}x^5 + \cdots\right) + a_1 \left(x - \frac{1}{48}x^4 + \cdots\right) + \\ \left(\frac{1}{4}x^2 + \frac{1}{24}x^3 - \frac{1}{96}x^4 - \frac{1}{160}x^5 + \cdots\right)$$

⇒ Note that, the third series is the particular solution, while the first and second series together represent the general solution of the associated homogeneous equation

$$(x^2 + 4)y'' + xy = 0$$

4. Find the general solution near $x = -1$ of:

$$y'' + xy' + (2x - 1)y = 0$$

SOLUTION

⇒ We make the substitution $t = x - (-1) = x + 1 \Rightarrow t = x + 1$.

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \text{ and } \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} \cdot \frac{dt}{dx} = \frac{d^2y}{dt^2}$$

⇒ From $t = x + 1$, we have $x = (t - 1)$.

⇒ Substituting these results into the differential equation, we get;

$$\frac{d^2y}{dt^2} + (t - 1)\frac{dy}{dt} + (2t - 3)y = 0$$

⇒ The solution for this equation is found near $t = 0$.

⇒ Here $p(t) = t - 1$ and $q(t) = 2t - 3$ are both polynomials, hence every point in particular $t = 0$ is an ordinary point.

⇒ Substituting eqns (5) through (7) into the left side of the differential equation, we have;

$$\begin{aligned} & [2a_2 + 6a_3t + 12a_4t^2 + \dots + (n+2)(n+1)a_{n+2}t^n + \dots] + \\ & (t-1)[a_1 + 2a_2t + 3a_3t^2 + \dots + na_nt^{n-1} + (n+1)a_nt^n + \dots] + \\ & (2t-3)[a_0 + a_1t + a_2t^2 + \dots + a_{n-1}t^{n-1} + a_nt^n + \dots] = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & [2a_2 + 6a_3t + 12a_4t^2 + \dots + (n+2)(n+1)a_{n+2}t^n + \dots] + \\ & t[a_1 + 2a_2t + 3a_3t^2 + \dots + na_nt^{n-1} + (n+1)a_nt^n + \dots] \\ & - [a_1 + 2a_2t + 3a_3t^2 + \dots + na_nt^{n-1} + (n+1)a_nt^n + \dots] + \\ & 2t[a_0 + a_1t + a_2t^2 + \dots + a_{n-1}t^{n-1} + a_nt^n + \dots] \\ & - 3[a_0 + a_1t + a_2t^2 + \dots + a_{n-1}t^{n-1} + a_nt^n + \dots] = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & (2a_2 - a_1 - 3a_0) + (6a_3 + a_1 - 2a_2 + 2a_0 - 3a_1)t + \dots \\ & + [(n+2)(n+1)a_{n+2} + na_n - (n+1)a_{n+1} + 2a_{n-1} - 3a_n]t^n + \dots = 0 + 0t + \dots + 0t^n + \dots \end{aligned}$$

⇒ Equating each coefficient to zero, we obtain:

$$2a_2 - a_1 - 3a_0 = 0, \quad 6a_3 - 2a_2 - 2a_1 + a_0 = 0, \quad \dots \quad (*)$$

⇒ In general:

$$(n+2)(n+1)a_{n+2} - (n+1)a_{n+1} + (n-3)a_n + 2a_{n-1} = 0$$

$$\Rightarrow a_{n+2} = \frac{1}{(n+2)}a_{n+1} - \frac{(n-3)}{(n+2)(n+1)}a_n - \frac{2}{(n+2)(n+1)}a_{n-1}$$

⇒ This recurrence relation is not valid for $n = 0$, because a_{n-1} is an undefined quantity.

⇒ To obtain an equation for $n = 0$, we use the first equation in (*) which gives:

$$a_2 = \frac{1}{2}a_1 + \frac{3}{2}a_0$$

⇒ Now evaluating recurrence formula for successive integer values of $n = 1, 2, 3, \dots$, we find that:

$$a_3 = \frac{1}{3}a_2 + \frac{1}{3}a_1 - \frac{1}{3}a_0 = \frac{1}{3}\left(\frac{1}{2}a_1 + \frac{3}{2}a_0\right) + \frac{1}{3}a_1 - \frac{1}{3}a_0 = \frac{1}{2}a_1 + \frac{1}{6}a_0,$$

$$a_4 = \frac{1}{4}a_3 + \frac{1}{12}a_2 - \frac{1}{6}a_1 = \frac{1}{4}\left(\frac{1}{2}a_1 + \frac{1}{6}a_0\right) + \frac{1}{12}\left(\frac{1}{2}a_1 + \frac{3}{2}a_0\right) - \frac{1}{6}a_1 = \frac{1}{6}a_0, \dots$$

⇒ Thus from $y(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + \dots$, we get:

$$y(t) = a_0 + a_1t + \left(\frac{1}{2}a_1 + \frac{3}{2}a_0\right)t^2 + \left(\frac{1}{2}a_1 + \frac{1}{6}a_0\right)t^3 + \left(\frac{1}{6}a_0\right)t^4 + \dots$$

$$\Rightarrow y(t) = a_0 \left(1 + \frac{3}{2}t^2 + \frac{1}{6}t^3 + \frac{1}{6}t^4 + \dots\right) + a_1 \left(t + \frac{1}{2}t^2 + \frac{1}{2}t^3 + 0t^4 + \dots\right)$$

⇒ Substituting back $t = (x+1)$, we obtain as the solution to the original problem:

$$y(x) = a_0 \left(1 + \frac{3}{2}(x+1)^2 + \frac{1}{6}(x+1)^3 + \frac{1}{6}(x+1)^4 + \dots\right) +$$

$$a_1 \left((x+1) + \frac{1}{2}(x+1)^2 + \frac{1}{2}(x+1)^3 + \dots\right)$$

5. Solve the initial-value problem:

$$y'' + xy' + (2x-1)y = 0; \quad y(-1) = 2, \quad y'(-1) = -2$$

SOLUTION

⇒ Since the initial conditions are prescribed at $x = -1$, it is advantageous to obtain the general solution to the differential equation near $x = -1$.

⇒ This has already been done in Problem 4 above, and the solution is:

$$y(x) = a_0 \left(1 + \frac{3}{2}(x+1)^2 + \frac{1}{6}(x+1)^3 + \frac{1}{6}(x+1)^4 + \dots\right) +$$

$$a_1 \left((x+1) + \frac{1}{2}(x+1)^2 + \frac{1}{2}(x+1)^3 + \dots\right)$$

$\Rightarrow$ Now applying the initial conditions, we have;

$$\begin{aligned} a_0 \left(1 + \frac{3}{2}(-1+1)^2 + \frac{1}{6}(-1+1)^3 + \frac{1}{6}(-1+1)^4 + \dots \right) + \\ a_1 \left((-1+1) + \frac{1}{2}(-1+1)^2 + \frac{1}{2}(-1+1)^3 + \dots \right) = 2 \\ \Rightarrow a_0 = 2 \end{aligned}$$

$$\begin{aligned} y'(x) &= a_0 \left(3(x+1) + \frac{1}{2}(x+1)^2 + \frac{2}{3}(x+1)^3 + \dots \right) + \\ &\quad a_1 \left(1 + (x+1) + \frac{3}{2}(x+1)^2 + \dots \right) \\ \Rightarrow a_0 \left(3(-1+1) + \frac{1}{2}(-1+1)^2 + \frac{2}{3}(-1+1)^3 + \dots \right) + \\ &\quad a_1 \left(1 + (-1+1) + \frac{3}{2}(-1+1)^2 + \dots \right) = -2 \\ \Rightarrow a_1 &= -2 \end{aligned}$$

$\Rightarrow$ Hence, $a_0 = 2$ and $a_1 = -2$, therefore the solution is:

$$\begin{aligned} y(x) &= 2 \left(1 + \frac{3}{2}(x+1)^2 + \frac{1}{6}(x+1)^3 + \frac{1}{6}(x+1)^4 + \dots \right) + \\ &\quad (-2) \left((x+1) + \frac{1}{2}(x+1)^2 + \frac{1}{2}(x+1)^3 + \dots \right) \\ \Rightarrow y(x) &= 2 - 2(x+1) + 2(x+1)^2 - \frac{2}{3}(x+1)^3 + \frac{1}{3}(x+1)^4 + \dots \end{aligned}$$

2.1.2.2: Recurrence Formula by Using Sigma/Summation Notation

CONCEPTS:

- 1) It is best to use summation notation, because it is more concise and also leads to the recurrence formula/recurrence relation.
- 2) The recurrence relation is important because it permits the calculation of as many coefficients of the series as we desire.
- 3) The method may be outlined as followed:
 - (i) Write the differential equation in the standard form $y'' + p(x)y' + q(x)y = 0$ to identify $p(x)$ and $q(x)$.
 - (ii) Choose an expansion point x_0 at which $p(x)$ and $q(x)$ are analytic. If initial conditions are given at some point it is suggested that, that point be used as x_0 .
 - (iii) Seeking $y(x)$ in the form of a power series about x_0 , put that form into the differential equation.

- (iv) By changing dummy indices of summation and lower summation limits, as necessary, obtain a form where each summation has the **Same Exponent** on $(x - x_0)$ and the **Same Summation Limits**.
- (v) Combine the sum into a single sum.
- (vi) Set the coefficient of $(x - x_0)^n$ within that sum to zero, that step gives the recursion formula.
- (vii) From the recursion formula, obtain as many of the coefficients as desired and hence the solution form

$$y(x) = Ay_1(x) + By_2(x)$$

where A and B are the arbitrary constants and $y_1(x)$ and $y_2(x)$ are the **Linearly Independent Power Series**.

- 4) Before illustrating the approach by means of an example, we first find expressions for derivatives $y'(x)$ and $y''(x)$ that will be needed in the calculation.
- 5) Writing out the first few terms of $y(x)$ gives:

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots = \sum_{n=0}^{\infty} a_nx^n \quad (1)$$

- 6) Differentiating (1) term by term with respect to x , we get:

$$y'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots = \sum_{n=1}^{\infty} na_nx^{n-1} \quad (2)$$

and after a further differentiation, we have;

$$y''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \cdots = \sum_{n=2}^{\infty} n(n-1)a_nx^{n-2} \quad (3)$$

- 7) In what is to follow, it will be important to remember that the summation in (2) starts at $n = 1$, whereas the summation in (3) starts at $n = 2$.

EXAMPLES

- 1. Solve $y'' + y = 0$ by the power series method.

SOLUTION

$\Rightarrow$ We can choose the point of expansion x_0 as any point at which both $p(x)$ and $q(x)$ are analytic.

$\Rightarrow$ Here $p(x) = 0$ and $q(x) = 1$ are analytic for all x , so we can choose the point of expansion x_0 to be whatever we like.

$\Rightarrow$ Let $x_0 = 0$, for simplicity, then all solutions can be found in the form:

$$y(x) = \sum_{n=0}^{\infty} a_nx^n \quad (1)$$

⇒ Then

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad (2)$$

and

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \quad (3)$$

⇒ Thus the differential equation becomes:

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^{n-1} = 0 \quad (4)$$

⇒ We would like to combine the two sums, but to do that, we need the exponents of x to be the same as x^n .

⇒ To have the same exponents, let us set $n-2 = m$ in the first sum. With $n-2 = m$, $n = \infty$ corresponds to $m = \infty$ and $n = 2$ corresponds to $m = 0$, thus (4) becomes:

$$\sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m + \sum_{n=0}^{\infty} a_n x^n = 0 \quad (5)$$

⇒ Next, let $m = n$ in the first sum in (5), we get:

$$\begin{aligned} \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + a_n] x^n &= 0 \end{aligned} \quad (6)$$

⇒ Finally, it follows that each coefficient in (6) must be zero. That is:

$$(n+2)(n+1) a_{n+2} + a_n = 0; \quad \text{for } n = 0, 1, 2, 3, \dots \quad (7)$$

⇒ Equation (7) is known as a **Recursion Formula**.

⇒ Specifically:

$$a_{n+2} = -\frac{1}{(n+2)(n+1)} a_n; \quad \text{for } n = 0, 1, 2, 3, \dots$$

⇒ So that:

$$\begin{aligned} a_2 &= -\frac{1}{2} a_0 = -\frac{1}{2!} a_0, \\ a_3 &= -\frac{1}{6} a_1 = -\frac{1}{3!} a_1, \\ a_4 &= -\frac{1}{12} a_2 = \frac{1}{24} a_0 = \frac{1}{4!} a_0, \\ a_5 &= -\frac{1}{20} a_3 = \frac{1}{120} a_1 = \frac{1}{5!} a_1, \dots \end{aligned}$$

⇒ Where a_0 and a_1 remain arbitrary.

$\Rightarrow$ Put these results into (1) gives:

$$y(x) = a_0 + a_1x - \frac{1}{2!}a_0x^2 - \frac{1}{3!}a_1x^3 + \frac{1}{4!}a_0x^4 + \frac{1}{5!}a_1x^5 + \dots$$

$$\Rightarrow y(x) = a_0 \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - + \dots \right) + a_1 \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - + \dots \right)$$

$\Rightarrow$ But $1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - + \dots = \cos x$ and $x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - + \dots = \sin x$

$\Rightarrow$ Therefore, $y(x) = a_0 \cos x + a_1 \sin x$.

2. Find the recurrence relation that must be satisfied by coefficients in the series solution of:

$$y'' + 2xy' + (1 + x^2)y = 0$$

when the expansion is about the origin.

Solve the initial value problem for the differential equation given that $y(0) = 3$ and $y'(0) = -1$.

SOLUTION

$\Rightarrow$ Given $y'' + 2xy' + (1 + x^2)y = 0$. Thus substitute:

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\text{and } y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

into the differential equation, we get:

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + 2x \sum_{n=1}^{\infty} n a_n x^{n-1} + (1 + x^2) \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

$\Rightarrow$ For the sum $\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$, let $n-2 = m \Rightarrow n = m+2$, thus we have;

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{m=0}^{\infty} (m+2)(m+1) a_{m+2} x^m$$

$$= \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

$\Rightarrow$ For the sum $\sum_{n=0}^{\infty} a_n x^{n+2}$, let $n+2 = m \Rightarrow n = m-2$, thus we have:

$$\sum_{n=0}^{\infty} a_n x^{n+2} = \sum_{m=2}^{\infty} a_{m-2} x^m = \sum_{n=2}^{\infty} a_{n-2} x^n$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=1}^{\infty} 2na_nx^n + \sum_{n=0}^{\infty} a_nx^n + \sum_{n=2}^{\infty} a_{n-2}x^n = 0$$

$\Rightarrow$ But $\sum_{n=1}^{\infty} 2na_nx^n = \sum_{n=0}^{\infty} 2na_nx^n$ and $\sum_{n=2}^{\infty} a_{n-2}x^n = \sum_{n=0}^{\infty} a_{n-2}x^n$ with $a_{-2} = 0$ and $a_{-1} = 0$.

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=0}^{\infty} 2na_nx^n + \sum_{n=0}^{\infty} a_nx^n + \sum_{n=0}^{\infty} a_{n-2}x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2na_n + a_n + a_{n-2}]x^n = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} + (2n+1)a_n + a_{n-2} = 0$$

$$\Rightarrow a_{n+2} = -\frac{(2n+1)}{(n+2)(n+1)}a_n - \frac{1}{(n+2)(n+1)}a_{n-2}; \text{ for } n = 0, 1, 2, \dots \text{ with } a_{-2} = 0 \text{ and } a_{-1} = 0$$

$\Rightarrow$ This equation for a_{n+2} is the recurrence relation.

$\Rightarrow$ Now setting $n = 0, 1, 2, 3, \dots$, with $a_{-2} = 0$ and $a_{-1} = 0$, we have:

$$a_2 = -\frac{1}{2}a_0,$$

$$a_3 = -\frac{1}{2}a_1,$$

$$a_4 = -\frac{5}{12}a_2 - \frac{1}{12}a_0 = -\frac{5}{12}\left(-\frac{1}{2}a_0\right) - \frac{1}{12}a_0 = \frac{1}{8}a_0,$$

$$a_5 = -\frac{7}{20}a_3 - \frac{1}{20}a_1 = -\frac{7}{20}\left(-\frac{1}{2}a_1\right) - \frac{1}{20}a_1 = \frac{1}{8}a_1$$

$\Rightarrow$ Continuing this process generates the coefficients:

$$a_6 = -\frac{1}{48}a_0, \quad a_7 = -\frac{1}{48}a_1, \quad a_8 = \frac{1}{384}a_0, \quad a_9 = \frac{1}{384}a_1, \dots$$

$\Rightarrow$ Put these coefficients into the power series $y(x) = \sum_{n=0}^{\infty} a_n x^n$ and grouping terms, gives:

$$y(x) = a_0 \left(1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \frac{1}{384}x^8 - + \dots \right) \\ + a_1 \left(x - \frac{1}{2}x^3 + \frac{1}{8}x^5 - \frac{1}{48}x^7 + \frac{1}{384}x^9 - + \dots \right)$$

$\Rightarrow$ To solve the initial value problem, the constants a_0 and a_1 must be chosen such that:

$$y(0) = 3 \quad \text{and} \quad y'(0) = -1$$

$$\begin{aligned} y(0) &= a_0 \left(1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \frac{1}{384}x^8 - + \dots \right) \\ &+ a_1 \left(x - \frac{1}{2}x^3 + \frac{1}{8}x^5 - \frac{1}{48}x^7 + \frac{1}{384}x^9 - + \dots \right) = 3 \end{aligned}$$

$$\Rightarrow a_0 + 0 = 3 \Rightarrow a_0 = 3$$

$$\begin{aligned} y'(x) &= a_0 \left(-x + \frac{1}{2}x^3 - \frac{1}{8}x^5 + \frac{1}{48}x^7 - + \dots \right) \\ &+ a_1 \left(1 - \frac{3}{2}x^2 + \frac{5}{8}x^4 - \frac{7}{48}x^6 + \frac{9}{384}x^8 - + \dots \right) \end{aligned}$$

$$\begin{aligned} y'(0) &= a_0 \left(-x + \frac{1}{2}x^3 - \frac{1}{8}x^5 + \frac{1}{48}x^7 - + \dots \right) \\ &+ a_1 \left(1 - \frac{3}{2}x^2 + \frac{5}{8}x^4 - \frac{7}{48}x^6 + \frac{9}{384}x^8 - + \dots \right) = -1 \end{aligned}$$

$$\Rightarrow 0 + a_1 = -1 \Rightarrow a_1 = -1$$

$\Rightarrow$ Thus $a_0 = 3$ and $a_1 = -1$, therefore the solution to the initial value problem is:

$$\begin{aligned} y(x) &= 3 \left(1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \frac{1}{384}x^8 - + \dots \right) \\ &- \left(x - \frac{1}{2}x^3 + \frac{1}{8}x^5 - \frac{1}{48}x^7 + \frac{1}{384}x^9 - + \dots \right) \end{aligned}$$

$$y(x) = 3 - x - \frac{3}{2}x^2 + \frac{1}{2}x^3 + \frac{3}{8}x^4 - \frac{1}{8}x^5 - \frac{1}{16}x^6 + \frac{1}{48}x^7 + \frac{1}{128}x^8 - \frac{1}{384}x^9 - \dots$$